Manufacturing Cost and Profitability Analysis

1. Problem statement

Develop a Python program to calculate manufacturing costs, including raw materials, labour, wastage, and machine costs. Determine whether the product achieves the required profit target.

2. Objectives

- Calculate material and labour costs.
- Include wastage and machine costs.
- Calculate total production cost.
- Determine profit and profit percentage.
- Check the profit target.

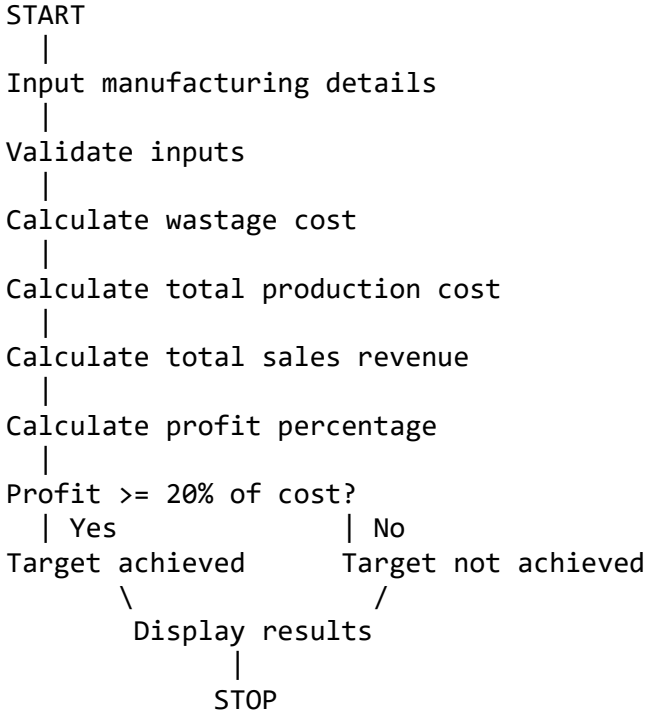
3. Assumptions

- Material cost and labour cost are entered by the user.
- Wastage is 5% of material cost.
- Machine cost is entered by the user.
- Selling price and production quantity are entered by the user.
- Target profit is 20% of total production cost.

4. Algorithm

1. Input material cost, labour cost, machine cost, selling price per unit, and quantity.
2. Validate the inputs.
3. Calculate wastage cost.
4. Calculate total production cost.
5. Calculate total sales revenue.
6. Calculate profit and profit percentage.
7. Compare profit percentage with the target.
8. Display the results.

5. Flowchart



6. Pseudocode

```
START
INPUT material, labour, machine, price, quantity

wastage = material * 5 / 100
cost = material + labour + machine + wastage
revenue = price * quantity
profit = revenue - cost
profit_percentage = profit / cost * 100

IF profit_percentage >= 20
    PRINT "Target achieved"
ELSE
    PRINT "Target not achieved"
END IF

PRINT cost, revenue, profit
STOP
```

7. Python source code

```
material = float(input("Enter material cost: "))
labour = float(input("Enter labour cost: "))
machine = float(input("Enter machine cost: "))
price = float(input("Enter selling price per unit: "))
quantity = int(input("Enter production quantity: "))

if min(material, labour, machine) < 0 or price < 0 or quantity <= 0:
    print("Invalid input")
else:
    wastage = material * 0.05
    total_cost = material + labour + machine + wastage
    revenue = price * quantity
    profit = revenue - total_cost

    if total_cost > 0:
        percentage = profit / total_cost * 100
    else:
        percentage = 0

    print("Wastage cost:", round(wastage, 2))
    print("Total production cost:", round(total_cost, 2))
    print("Sales revenue:", round(revenue, 2))
    print("Profit:", round(profit, 2))
    print("Profit percentage:", round(percentage, 2), "%")

    if percentage >= 20:
        print("Profit target achieved")
```

```
    else:
        print("Profit target not achieved")
```

8. Test cases

Test	Material	Labour	Machine	Price/unit	Quantity	Expected result
1	₹5,000	₹2,000	₹1,000	₹100	100	Target achieved
2	₹5,000	₹2,000	₹1,000	₹70	100	Target not achieved
3	₹1,000	₹500	₹200	₹30	100	Target achieved
4	-₹1,000	₹500	₹200	₹30	100	Invalid input